# Panskura Banamali College (Autonomous)

**M.Sc. 4th Semester Examination - 2021 (Online Mode)**

**Subject: Computer Science** **Paper: COS-401**

**Paper Name: Artificial Intelligence**

**F.M.: 40** **Time : 2 Hours**

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**A. Answer any two questions:** **( 2 x 4 = 8)**

1. What is Backward Chaining?
2. What is Depth First Search? Explain it.
3. What is Unsupervised Learning?

**B. Answer any two questions:** **( 2 x 8 = 16)**

1. Difference between Simulated annealing and Genetic Algorithm?
2. Write down the properties of Best First Search algorithm.
3. Explain Bayesian Reasoning in detail. Find the limitation of Bayesian Reasoning.

**C. Answer any one question:** **( 1x 16 =16)**

1. Prove that for any $n^*$ on an optimal path $P^*_{\gamma-1}$ always satisfies $f^*(n^*) = c^*$.
2. Prove by resolution theorem that the goal is derivable from the knowledge base

i. The humidity is high v the sky is cloudy ii. If the sky is cloudy then it will rain

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